

Kirk's Grafana Learning Lab

Step-by-step progress guide

Ubuntu Server 24.04 LTS • VMware Workstation • Remote Desktop • Grafana

Prepared for: Kirk Biggs

Role: Process Technician

Progress date: July 24, 2026

This guide records the lab work completed so far and provides a repeatable reference for rebuilding the environment.

Contents

Contents	2
1. Build the Ubuntu virtual machine	3
2. Prepare Ubuntu and enable remote desktop	4
3. Install and start Grafana	5
4. Open Grafana and confirm the installation	6
5. Current status and next milestone	7

How to use this guide

Completed work is marked in green. Commands are shown in dark boxes. Replace values inside angle brackets with the values from your own lab.

Lab at a glance

Layer	What is in place	Status
Host	Windows PC running VMware Workstation	Complete
Guest	Ubuntu Server 24.04 LTS virtual machine	Complete
Access	Graphical remote desktop connection	Complete
Monitoring	Grafana server and web interface	Complete
Build next	First data source and dashboard	Next

1. Build the Ubuntu virtual machine

The foundation of the lab is an Ubuntu Server virtual machine hosted in VMware Workstation. This keeps the Grafana environment isolated, repeatable, and easy to back up.

1 Download Ubuntu Server 24.04 LTS

COMPLETED

You downloaded the Ubuntu Server 24.04 LTS ISO image. The ISO is the installer media used by VMware to create the Linux server.

- ✓ Ubuntu Server 24.04 LTS ISO downloaded
- ✓ ISO kept on the Windows host for use by VMware

2 Create the virtual machine in VMware Workstation

COMPLETED

A new VM was created and connected to the Ubuntu ISO. Typical lab settings are shown below; the exact values may differ from your machine.

Setting	Lab value / recommendation
Guest OS	Linux - Ubuntu 64-bit
Processors	2 virtual CPUs
Memory	4 GB RAM
Disk	25 GB or more
Network	NAT for simple internet access
Installer	Ubuntu Server 24.04 LTS ISO

3 Install and start Ubuntu Server

COMPLETED

You completed the Ubuntu installer, created your Linux user, and reached a working server. After installation, the ISO can be disconnected from the VM so the server boots from its virtual disk.

- ✓ Ubuntu boots successfully
- ✓ Linux username and password work
- ✓ VM has network access

Record your lab details

VM name: _____ Linux user: _____
Server IP: _____ VMware network mode: _____

2. Prepare Ubuntu and enable remote desktop

4 Update the operating system

COMPLETED

The package lists and installed software were updated before adding Grafana.

```
sudo apt update
sudo apt upgrade -y
```

If Ubuntu reports that a reboot is required, restart the VM and reconnect before continuing.

```
sudo reboot
```

5 Install a desktop and remote desktop service

COMPLETED

Because Ubuntu Server starts without a full graphical desktop, a lightweight desktop environment and the XRDP service were installed so the VM can be controlled from Windows Remote Desktop Connection.

```
sudo apt install -y xubuntu-desktop xrdp
sudo systemctl enable --now xrdp
sudo adduser xrdp ssl-cert
```

Confirm that XRDP is running:

```
systemctl status xrdp --no-pager
```

Expected result

The service status should show **active (running)**. Press **Q** if the status view remains open.

Find the VM's IP address:

```
hostname -I
```

6 Connect from Windows

COMPLETED

On the Windows host, you opened **Remote Desktop Connection** (mstsc), entered the Ubuntu VM's IP address, and signed in with the Linux username and password.

- ✓ Remote desktop window opens
- ✓ Ubuntu desktop is visible
- ✓ Keyboard and mouse work inside the VM

Connection tip

Use the VM's current IPv4 address. With VMware NAT, the address can change after a restart. Run **hostname -I** again if the old address stops working.

3. Install and start Grafana

7 Add the Grafana package repository

COMPLETED

Grafana was installed from Grafana's official APT repository. The commands below are a clean reference for repeating that work on Ubuntu 24.04.

```
sudo apt-get install -y apt-transport-https software-properties-common wget
sudo mkdir -p /etc/apt/keyrings
wget -q -O - https://apt.grafana.com/gpg.key | \
gpg --dearmor | sudo tee /etc/apt/keyrings/grafana.gpg > /dev/null

echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] \
https://apt.grafana.com stable main" | \
sudo tee /etc/apt/sources.list.d/grafana.list
```

8 Install Grafana

COMPLETED

```
sudo apt update
sudo apt install -y grafana
```

9 Enable and start the Grafana service

COMPLETED

```
sudo systemctl daemon-reload
sudo systemctl enable --now grafana-server
systemctl status grafana-server --no-pager
```

Expected result

The Grafana service should show **active (running)**. Enabling the service makes Grafana start automatically when Ubuntu boots.

If Ubuntu's firewall is enabled, allow Grafana's default TCP port:

```
sudo ufw allow 3000/tcp
sudo ufw status
```

4. Open Grafana and confirm the installation

10 Open the Grafana web interface

COMPLETED

From a browser on the Windows host or inside the Ubuntu desktop, open the server's address on port 3000.

```
http://<server-ip>:3000
```

Example only: if the server address is 192.168.100.25, open **http://192.168.100.25:3000**.

11 Sign in and secure the administrator account

COMPLETED

For a new Grafana installation, the initial sign-in is normally **admin / admin**. Grafana then asks for a new password. The password should be unique to this lab and stored securely.

- ✓ Grafana login page loads
- ✓ Administrator password changed
- ✓ Grafana home page or welcome screen appears

Quick health checks

Check	Command or test	Good result
Service	systemctl is-active grafana-server	active
Listening port	ss -ltn grep 3000	Port 3000 listed
Local HTTP	curl -I http://localhost:3000/login	HTTP response
Windows browser	http://<server-ip>:3000	Grafana login/home

Troubleshooting sequence

If the page does not open: (1) verify the service, (2) verify port 3000 is listening, (3) confirm the server IP, (4) check UFW, and (5) confirm the VMware network adapter is connected.

5. Current status and next milestone

Your lab platform is ready. The Ubuntu VM is running, remote access is working, and Grafana is available. The next phase is to connect a data source and build the first dashboard.

Completed so far

- ✓ Downloaded Ubuntu Server 24.04 LTS
- ✓ Created and installed the Ubuntu VM in VMware Workstation
- ✓ Updated Ubuntu
- ✓ Configured graphical remote desktop access
- ✓ Installed Grafana
- ✓ Enabled and started the Grafana service
- ✓ Reached the Grafana web interface

Next: build the first dashboard

1	Choose a data source	Start with Grafana's built-in TestData source for practice, or install Prometheus for real host metrics.
2	Add the data source	In Grafana, open Connections > Data sources and complete the connection settings.
3	Create a dashboard	Choose Dashboards > New > New dashboard > Add visualization.
4	Build the first panel	Select the data source, create a query, choose a visualization, and save the panel.
5	Save and document	Name the dashboard, record what each panel shows, and take a VM snapshot.

Recommended checkpoint

Before adding more software, shut down Ubuntu cleanly and create a VMware snapshot named **Grafana Installed - Clean**. This gives you a fast recovery point.

Lab notes

Grafana URL: _____

VM snapshot: _____

Next data source: _____

Notes: _____

[End of progress guide • Ready for the first Grafana dashboard](#)